

IBM DATA SCIENCE COURSE

Coursera Link – [IBM Data Science Professional Certificate | Coursera](#)

EDX Link – [IBM Data Science Professional Certificate](#)

INTRODUCTION TO DATA SCIENCE

- What is Data Science?
- Tools and Skillset?
- Methodologies?
- Role of Data Science in real life?

BASIC TOOLS FOR DATA SCIENCE

- Basic Overview of R, Python, SQL
- Jupyter Notebook
- Python Libraries Overview
- Github
- Google Cloud

PYTHON FOR DATA SCIENCE

MODULE 1 – INTRODUCTION TO PYTHON

- Data Types, Operators, Input/Output
- Python Data Structures
- Control Statements and Logic Flow (if/else)
- Loops
- Functions
- Strings
- OOPs

MODULE 2 – FILING

- Open and Closing File
- Read/Write Append
- With keyword
- Loading and Reading data with Pandas

MODULE 3 – INTRO TO AI

- Agents and its Types
- Basic Searches
- Advances Searches (heuristic)
- Local Searches
- CSP
- Game Theory
- Probability
- Intro to Pandas/Numpy/Matplotlib

MODULE 4 – WEB SCRAPING and APIs

- URL parsing
- Storing Data with BeautifulSoup
- Finding Tags
- API calls
- Stock Analysis Project

SQL FOR DATA SCIENCE

- Basics of SQL
- DDL and DML
- Grouping, Aggregation, Sub-queries
- Tables and Relations
- Joins
- Views
- PL/SQL
- Triggers
- ERD
- Intro to MongoDB
- Accessing Databases with Python

DATA ANALYSIS WITH PYTHON

MODULE 1 – DATA WRANGLING

- Importing Dataset and Packages
- Understanding Data
- Pre-processing – Data Cleaning and Normalization

MODULE 2 – EDA

- Basic Statistics and Sorting
- Grouping and Functions
- Summary Tables
- Data Frames
- Correlations – Analysis and Heatmap
- Predictions (CrossTab)
- Plotting (Graphs)

DATA VISUALIZATION

- Matplotlib and Seaborn
- Types of Plots: Area, Histogram, Bar, Pie, Scatter, Box, Bubble
- Advanced Plots: Waffle Charts, Word Clouds
- Plotly
- Pair Plots and Violin Plots
- Seaborn and Regression Plots
- Maps and Geospatial Data: Folium, Maps with Markers, Choropleth Maps

MACHINE LEARNING

- Learning and its Types
- EDA (done)
- Scikit Learn library: Train/Test Split, Confusion Matrix, F1, Feature Scaling
- Pipeline for Model Development
- Ensemble Methods
- Supervised Models: Linear Regression and Types, SVM, Decision Tree, KNN, RandomForest
- Unsupervised Models: K-means Clustering, Hierarchical Clustering, Density-based Clustering
- Reinforcement Learning: Q-star Algorithm
- Model Evaluation: Overfitting, Underfitting, Bias, Accuracy and other evaluation metrics
- Recommender Systems: Content-based recommender systems, Collaborative Filtering
- Real World Example: House Price Prediction

CAPSTONE PROJECT

- EDA with Python and SQL along with visualization

- Dashboards: PowerBI, PlottyDash
- Predictive Analysis
- Presenting Data Driver Insights

LEVERAGING GENERATIVE AI

- Gen AI's Impact across industries and its use case in data science
- Types of GenAI Models
- GenAI for Data Augmentation, Querying, Preparation
- GenAI tools for Data Analysis, Visualization, Model Development and EDA
- Final Project